

G Anvil

A two-voice analogue percussion module - one big kick and one snare / metal voice - with a 16-step sequencer.



Overview

G Anvil is a dual-voice analogue percussion module in the spirit of the heavy-duty German drum modules that built industrial music and techno. There are no samples: both voices are synthesised as they play.

The KICK is a bridged-T style oscillator with an exponential pitch sweep, a click for the attack and a hard, lopsided drive. The SNARE / METAL voice uses two detuned square oscillators and white noise through a resonant multi-mode filter: with METAL at 0 it is a fat snare body, turned up it becomes an inharmonic clang with ring modulation. Each voice has its own drive, and BUS SAT can push the whole module into saturation.

A two-row, 16-step sequencer plays the two voices in time with REAPER's transport, and both voices also play from MIDI notes, with velocity.

Reach for it for

- Big, distorted techno and industrial kicks
- Long, booming dub and rumble kicks

- Metallic clanks, rimshots and noisy snares from one voice
- Gabber and hardcore kicks with the drive turned right up
- Quick two-voice patterns right inside REAPER, or layering under another drum machine

Loading and playing it

1. Insert G Anvil on a new track. It is an instrument: it makes its own sound and needs no audio input.
2. Make sure PLAY (in TRANSPORT) is lit, then press play in REAPER. The sequencer follows the project's tempo and position.
3. Click a voice's orange LED to hear it, choose a preset, and edit the steps in the grid.
4. To play it from MIDI, arm the track with a MIDI input and record monitoring on, or put a MIDI item on the track (see MIDI notes below). Switch PLAY off if you want MIDI only. The MIDI light in the header flashes when a drum note arrives.

Controls

The header

PRESETS	Opens the full preset list.
◀ / ▶	Step to the previous or next preset (wrapping round).
Preset display	Shows the preset number and name. Click it for the preset list too.
MIDI	Flashes when a drum note arrives.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

KICK

Click the LED at the top left to hear the kick.

PITCH	Where the kick's pitch lands after the sweep, 30 - 150 Hz (50 Hz by default).
SWEEP	How far above PITCH the sweep starts, 0 - 48 semitones (24 by default). 0 gives a flat, tonal boom; more gives a punchier drop.
S.TIME	How fast the pitch falls, 2 - 250 ms (28 ms by default). Short for a tight punch, long for a laser-like zap.
DECAY	How long the kick rings, 40 - 2500 ms (420 ms by default), measured to -60 dB.
WAVE	Morphs the oscillator from a pure sine (SINE) to a triangle (TRIANGLE), 0 - 100% (10% by default). More triangle adds harmonics and helps the kick cut through on small speakers.
CLICK	The level of the attack click, a short pulse and a burst of noise, 0 - 100% (35% by default).
C.TONE	The click's colour, 0 - 100% (50% by default): from a dark thud to a bright tick. Brighter clicks are also shorter.
DRIVE	A hard, lopsided analogue-style saturation, 0 - 100% (25% by default). Low settings thicken the kick; high settings flatten it into a square-edged distorted kick.
LEVEL	The kick's volume, 0 - 100% (85% by default).
PAN	Position, -100 (left) to +100 (right); CENTRE by default.

SNARE / METAL

Click the LED at the top left to hear the snare.

TUNE	The pitch of the two square oscillators, 80 - 1200 Hz (210 Hz by default).
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METAL	0 - 100% (20% by default). Spreads the two oscillators apart and adds ring modulation: from a fat, nearly unison snare body to the classic inharmonic metal clang. It also opens up the tone's brightness.
T.DECAY	The decay of the tone, 10 - 2000 ms (110 ms by default).
N.DECAY	The decay of the noise, 10 - 2000 ms (190 ms by default).
NOISE	The balance of tone and noise, from ALL TONE (0%) to ALL NOISE (100%); 60% by default.
SNAP	The attack, 0 - 100% (50% by default): a quick level jump and a pitch blip at the start of each hit.
CUTOFF	The filter frequency, 100 Hz - 16 kHz (7.5 kHz by default).
RESO	The filter's resonance, 0 - 100% (20% by default). The level is kept steady as you turn it up.
LP / BP / HP (MODE)	The filter type: low-pass (the default, for a full snare), band-pass (a focused, boxy or ringing sound) or high-pass (thin, bright and clangy).
DRIVE	The same hard saturation as the kick, 0 - 100% (15% by default).
LEVEL	The snare's volume, 0 - 100% (75% by default).
PAN	Position, -100 (left) to +100 (right); CENTRE by default.

MASTER

LEVEL	Output level, -24 to +12 dB (0 dB by default).
BUS SAT	Lopsided saturation on the whole module, 0 - 100% (10% by default). It glues the two voices together and adds weight; high settings crunch hard.
L / R meters	The output level, left and right.

SEQUENCER

Two rows of 16 sixteenth-note steps, KICK on top and SNARE below. A step is a rest, a hit (orange) or an accent (red, with a light bar). The step that is playing is highlighted, and the LED next to each row's name flashes as the voice plays.

Steps

Click a step to set or clear it; drag to paint along the rows. Shift-click sets an accent; right-click switches a step between hit and accent.

The pattern is saved with the project and with each preset.

TRANSPORT

PLAY

The sequencer on or off (on by default). When on, it plays whenever REAPER's transport is playing, locked to the project tempo and position.

Display

The step and the tempo while playing. With PLAY on and the transport stopped it reads WAITING FOR PLAY; with PLAY off, SEQ OFF - MIDI ONLY.

SWING

Delays the off-beat sixteenths, 0 - 75% (0% by default).

ACC (ACCENT)

How much louder accents are than normal hits, 0 - 100% (50% by default). At 0 every step plays at full level; at 100 normal hits are 45% quieter than accents.

CLEAR

Clears both rows.

The bottom strip

DRIFT

0 - 100% (30% by default). Every hit is detuned a little at random, and the tunings, the snare filter and the click tone wander slowly, like real analogue circuits. At 0% every hit is identical.

MIDI (CH)

The MIDI channel: OMNI (all channels, the default) or 1 - 16.

STEREO / MONO

STEREO keeps the two voices where PAN puts them; MONO folds the output to mono.

SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises. 0% by default.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper at -0.3 dBFS (no latency, the default), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). While the output clips, the CLIP key flashes red.

Knobs: drag up or down (Shift for fine moves); double-click or Ctrl-click to reset a knob to its default.

MIDI notes

G Anvil plays from these notes, with velocity: it sets the level, and softer hits are a little darker. Other MIDI passes straight through.

KICK	Notes 35 and 36 (C1)
SNARE	Notes 38 (D1), 39 and 40
RIM	Note 37: the snare voice, shorter and more tonal, with the filter opened up

Presets

Presets set both voices, BUS SAT, SWING, ACCENT and the pattern. MASTER LEVEL, DRIFT, PLAY, the MIDI channel, STEREO / MONO and the output stage are left as they are.

Init	The default kit: a clean, punchy kick on every beat and a snare on 2 and 4.
Industrial Thud	A longer, driven kick and a dark band-pass snare with lots of noise and a ringing filter.
Techno Rumble Kick	A long, low 1.4-second kick with an off-beat high-pass metal tick on the snare voice.

Classic Punch	A bright-clicking, tight kick and a clean snare, with a touch of swing.
Distorted Hardcore	A heavily driven kick with a deep sweep and a crunchy band-pass snare; BUS SAT at 40%.
Metal Clank	The snare voice at full METAL with a long tone: a clanging anvil hit, in a broken pattern.
Rimshot Tick	A short, clicky kick and a very short, high-pitched high-pass snare: a rimshot tick.
Deep Dub	A soft, low kick that rings for 1.8 seconds and a gentle, long snare, with swing.
Gabber	A fast-sweeping kick with full drive and lots of triangle, and BUS SAT at 50%.
Minimal Click	A short, clicky kick with almost no drive and a tiny high metal tick, with swing.
Warehouse Snare	A broken kick pattern with a big, noisy, driven band-pass snare.
EBM Stomp	A slow-sweeping driven kick with plenty of triangle and a thick low-pass snare.

Tips

- For a long techno kick, set DECAY over 1 second, SWEEP low and S.TIME around 30 - 60 ms, then add DRIVE to taste.
- The kick's DRIVE and BUS SAT add a lot of level: lower the voice LEVEL as you turn them up.
- Turn METAL up and NOISE down, with the filter in BP or HP, for cowbells, clanks and bell-like hits.
- Use note 37 from MIDI for rimshots between the snares without changing the snare settings.

- Set DRIFT to 0 for a machine-tight loop; leave some in for a more alive, hardware feel.

File: gfx-g-anvil.jsfx